JOURNAL OF BACTERIOLOGY, Dec. 2009, p. 7620–7622 Vol. 191, No. 24
0021-9193/09/$12.00 doi:10.1128/JB.01200-09
Copyright © 2009, American Society for Microbiology. All Rights Reserved.

Structural and Genetic Analysis of X-Ray Scattering by Spores of *Bacillus subtilis*[▽]

Xiangyun Qiu[1] and Peter Setlow[2]*

Laboratory of Physical and Structural Biology, Program in Physical Biology, National Institute of Child Health and Human Development, National Institutes of Health, Bethesda, Maryland 20892-0924,[1] and Department of Molecular, Microbial and Structural Biology, University of Connecticut Health Center, Farmington, Connecticut 06030-3305[2]

Received 4 September 2009/Accepted 5 October 2009

Dormant spores of *Bacillus subtilis* exhibit two prominent X-ray scattering peaks. These peaks persisted in spores lacking most α/β-type small, acid-soluble protein or the CotE protein responsible for assembly of much spore coat protein, but they were absent from spores of strains lacking the late sporulation-specific transcription factor GerE.

Spores of *Bacillus* species are formed in the process of sporulation and are metabolically dormant and extremely resistant to a variety of environmental stresses, including heat, radiation, toxic chemicals, and mechanical disruption (9, 23). Spore resistance is due to a number of factors, including (i) the protection of spore DNA by its saturation with a group of α/β-type small, acid-soluble spore proteins (SASP) that markedly alter DNA and chromosome structure; (ii) a low level of water in the spore's central region or core; (iii) a high level of pyridine-2,6-dicarboxylic acid (dipicolinic acid [DPA]) and its chelated divalent cations in the spore core (~25% of core dry weight); and (iv) the thick proteinaceous coat that is the outer layer of spores of most species and which is important in spore resistance to toxic chemicals, lytic enzymes, and predators (1, 3, 6, 12–14, 21, 23).

The spore coat of the best-studied sporeformer, *Bacillus subtilis*, is composed of ≥70 proteins unique to spores that are in a multilayered structure, including an outer coat and a lamellar inner coat, whose assembly is extremely complex and not well understood at the molecular level (3, 6, 12). There is also only rudimentary knowledge of the detailed overall structure of the spore coat. A number of years ago, X-ray diffraction studies showed that dormant *B. subtilis* spores contain one or more components that gave two relatively closely spaced X-ray scattering peaks (7, 8, 10, 11). The precise identity of the scattering component or components has not been established, although initial results strongly suggested that this component or components were in the spore coat (10). However, despite the substantial amount of knowledge accumulated recently on the composition and assembly of the *B. subtilis* spore coat (6, 12), there has been little progress in identifying the specific proteins that contribute to X-ray scattering of these spores. In this work, we examined X-ray scattering of wild-type *B. subtilis* spores, as well as spores of a number of isogenic mutants to begin the process of identifying the X-ray scattering component(s) in spores.

***B. subtilis* strains used and spore preparation and purification.** The *B. subtilis* strains used in this work were isogenic derivatives of strain PS832, a prototrophic derivative of strain 168, and were PS533 (22), carrying plasmid pUB110 encoding resistance to kanamycin; PS356 (termed $\alpha^-\beta^-$), with deletions of the *sspA* and *sspB* genes that encode ~85% of the spore's α/β-type SASP pool (16); PS3328 (5), with the coding sequence of the *cotE* gene encoding a coat morphogenetic protein replaced by a tetracycline resistance marker; PS4149 (5), with the majority of the coding sequence of the *gerE* gene encoding a transcription factor involved in the expression of many genes encoding coat proteins (*cot* genes) replaced by a spectinomycin resistance marker; and PS4150 (4), carrying the *cotE* and *gerE* mutations from strains PS3328 and PS4149, respectively.

B. subtilis spores were prepared on 2× Schaeffer's growth medium agar plates without antibiotics (18, 19). After 2 to 3 days at 37°C, plates were moved to 23°C for 2 to 4 days and spores were then harvested, purified, and stored as described previously (18, 19). All spores used in this work were free (>98%) of growing or sporulating cells, germinated spores, or cell debris, as observed by phase-contrast microscopy.

Measurement of X-ray scattering. X-ray scattering is the method of choice to detect and quantify periodic biomolecular assemblies. We used a fixed copper anode Diffractis 601 X-ray generator (Enraf-Nonius Service Corporation, Bohemia, NY) equipped with double-focusing mirrors (Charles Supper Company, Natick, MA). Lyophilized spore samples (5 mg) were sealed between two Mylar windows in a washer-like Teflon cell mounted on a temperature-controlled aluminum holder, and all measurements were at 20°C with an exposure time of 3 h. Two-dimensional scattering intensities from an image plate detector were analyzed with home-written MATLAB (The Math Works) codes.

X-ray scattering of spores of various *B. subtilis* strains. The X-ray scattering of wild-type *B. subtilis* spores showed two closely spaced circular rings, as indicated in Fig. 1A. The presence of these rings is clear evidence for some highly ordered structure in the spore. For quantitative analysis, we integrated the two-dimensional intensities (I) radially to obtain the scat-

* Corresponding author. Mailing address: Department of Molecular, Microbial and Structural Biology, University of Connecticut Health Center, Farmington, CT 06030-3305. Phone: (860) 679-2607. Fax: (860) 679-3408. E-mail: setlow@nso2.uchc.edu.

▽ Published ahead of print on 16 October 2009.

Downloaded from jb.asm.org at NAT INST OF HEALTH LIB on November 23, 2009

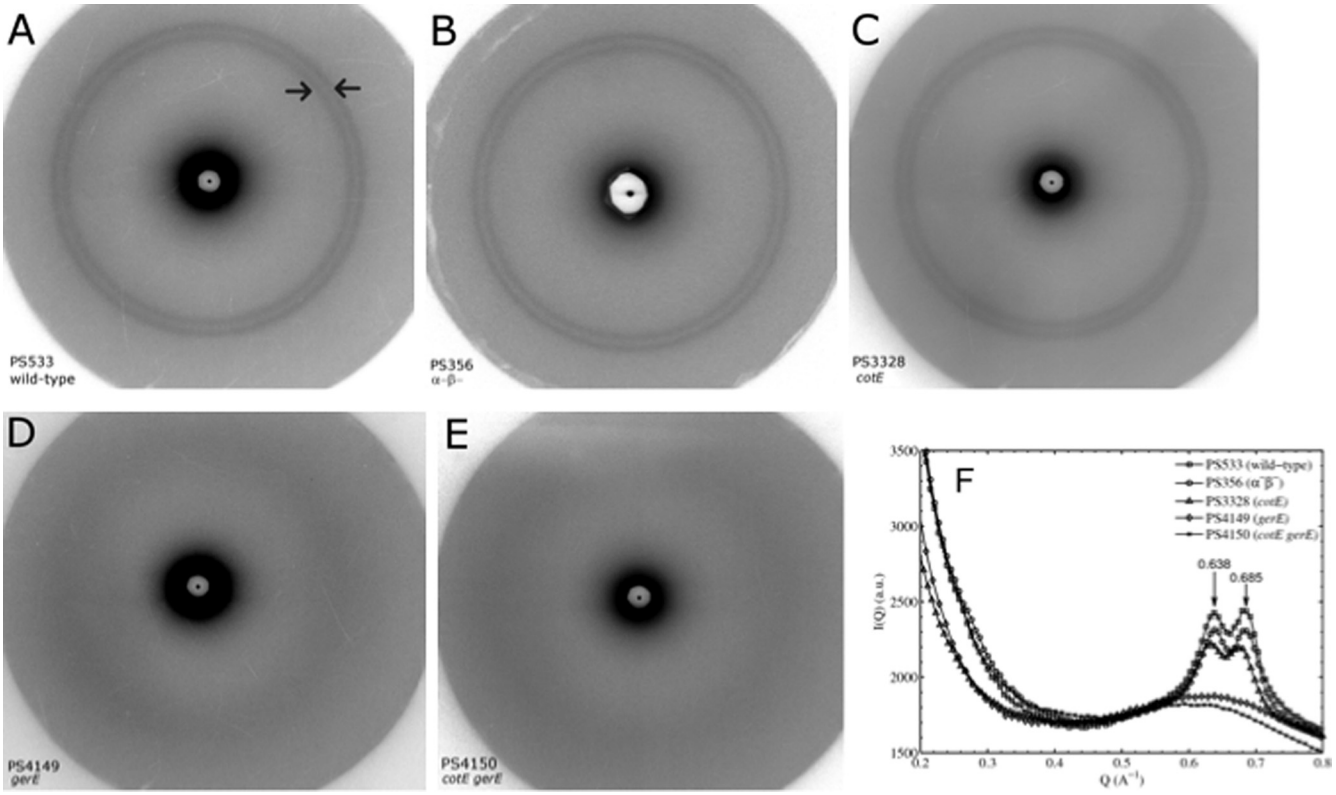


FIG. 1. X-ray scattering patterns of spores of various *B. subtilis* strains (A to E) and the radially integrated scattering intensities, $I(Q)$ (F). The strains are PS533 (wild type) (A), PS356 ($\alpha^-\beta^-$) (B), PS3328 (*cotE*) (C), PS4149 (*gerE*) (D), and PS4150 (*cotE gerE*) (E). Panel F shows the $I(Q)$ from panels A through E given in arbitrary units (a.u.).

tering profile $I(Q)$, where the scattering vector $Q = 4\pi\sin(\theta)/\lambda$, the scattering angle is 2θ, and λ is the X-ray wavelength. From the $I(Q)$ shown in Fig. 1F, it is straightforward to determine the positions of the double peak as 0.638 and 0.685 $Å^{-1}$, corresponding to spacings in real space of 9.85 and 9.17 Å, respectively. These values are consistent with previous observations (10).

The presence of the two scattering peaks noted above raises a number of interesting questions, including what spore layers give rise to the peaks and thus possess periodic structure, and what is the molecular identity of the components of these periodic structures? While previous work suggested that the scattering peaks are caused by one or more coat components (10), other possibilities have not been ruled out, including (i) the spore core's DPA, whose crystal structure has unit cell dimensions similar to the X-ray scattering periodicities seen in intact spores (20); or (ii) the novel spore DNA structure due to the DNA's saturation with α/β-type SASP. This latter saturation changes spore DNA structure from the normal B-helix to a structure with aspects of both A- and B-helical DNA and causes the spore chromosome to adopt a tightly packed toroidal structure (15, 21). To assess whether it is the α/β-type SASP-DNA complex that is responsible for the spore's X-ray scattering, we examined the scattering of $\alpha^-\beta^-$ spores that lack ~85% of total α/β-type SASP, since the DNA in these spores is (i) not present in a compact toroidal structure and (ii) most likely largely in a B-helical conformation (15, 21, 23). Strikingly, the X-ray scattering of $\alpha^-\beta^-$ spores was essentially identical to that of wild-type spores (compare Fig. 1B and A), thus ruling out α/β-type SASP-saturated DNA as a factor in the X-ray scattering peaks obtained with spores.

Given the result noted above, it seemed even more likely that one or more coat components are responsible for the X-ray scattering peaks obtained from intact spores. Consequently, we turned to the analysis of spores of strains with mutations that severely affected coat assembly and/or coat protein expression, yet still allowed spore formation and isolation. The first mutant spores tested lacked the CotE protein needed for normal spore coat assembly, and *cotE* spores lack a large number of normal coat proteins, the outer coat can no longer be seen in electron micrographs, and some likely inner coat proteins are also not assembled (6, 12). As found with $\alpha^-\beta^-$ spores, the X-ray scattering from *cotE* spores was again essentially identical to that from wild-type spores (compare Fig. 1C and A). However, note that the scattering peak positions were shifted to slightly lower Q values in the *cotE* spores (Fig. 1F), indicating that there is a small increase in the periodicities in these spores.

A second protein with a major role in coat formation is GerE, a transcription factor responsible for regulating the expression of a number of genes encoding coat proteins. Consequently, *gerE* spores lack or have very low levels of a number of coat proteins, including a few proteins that are assembled into spores even in the absence of CotE (4, 6, 12). The X-ray scattering measurements on *gerE* spores revealed that they gave no observable scattering peaks (compare Fig. 1D and A),

Downloaded from jb.asm.org at NAT INST OF HEALTH LIB on November 23, 2009

and similar results were obtained with *cotE gerE* spores (compare Fig. 1E and A). Both *gerE* and *cotE gerE* spores also have normal DPA and α/β-type SASP levels (5), further ruling out spore DNA as being responsible for the X-ray scattering peaks and also ruling out DPA, even though this compound may well be largely insoluble and perhaps crystalline in the spore core (23).

Conclusions. The combined structural and genetic analyses in the current work provide new insight into the structure and assembly of the spore coat as follows. (i) It is the spore coat alone that is responsible for the X-ray scattering peaks observed in dormant *B. subtilis* spores, as suggested previously from X-ray scattering of isolated spore coats (10). (ii) It is a coat protein or proteins whose expression or assembly into the spore coat is either directly or indirectly under the control of the GerE transcription factor that is essential for the X-ray scattering peaks. (iii) It is likely that these latter GerE-controlled proteins are in the inner spore coat, since *cotE* spores that still gave rise to the X-ray scattering peaks lack the outer coat, although *cotE* spores also lack some likely inner coat proteins as well (3, 6). In contrast, both the inner coat and the outer coat are largely absent from *gerE* and *cotE gerE* spores, as determined by electron and atomic force microscopy (5, 6, 17). Indeed, most *gerE* spores and essentially all *cotE gerE* spores have a surface almost devoid of any structural features, as seen by atomic force microscopy, in contrast to the surface ridges and bumps seen in wild-type and *cotE* spores (2, 5). However, *gerE* and *cotE gerE* spores do retain a thin layer of coat protein that is largely resistant to both proteases and solubilization by detergents (1, 5). This residual coat layer is thus likely to be significantly cross-linked, as are a number of proteins in the spore coat, although the precise mechanisms of coat protein cross-linking and the identities of all but a few cross-linked coat proteins are unknown (6). However, this residual, insoluble coat layer, which has been termed a "rind" (1, 13), is clearly not responsible for the structural periodicity of the intact dormant spore coat. It is worth noting that the layer-layer spacing of the lamellar inner coat of *B. subtilis* spores is in the order of few tens of nanometers, significantly larger than the ~1-nm spacings observed by X-ray scattering (3, 6). This leaves each single layer of the inner coat as the most likely candidate for the periodic structure in spores, although the underlying molecular origin of this periodic structure remains to be determined.

The genes regulated positively by GerE include a number that encode spore coat proteins, and at least some of these proteins are assembled in a CotE-independent manner (4, 6, 12). Thus, it should be possible to use this information to narrow down possible coat protein candidates for the component(s) giving rise to the X-ray scattering peaks of spores and the proteins generating the highly ordered structural features of the spore coat. Ultimately, this latter information could allow generation of this ordered spore coat structure in vitro to allow its more detailed structural analysis. Continued progress in the detailed molecular understanding of spore coat assembly and structure may then pave the way for the application of these principles in biotechnology and medicine.

This work was supported by a grant from the Army Research Office (P.S.) and by the Intramural Research Program of NIH, Program in Physical Biology, Eunice Kennedy Shriver National Institute of Child Health and Human Development (X.Q.).

REFERENCES

1. **Carroll, A. M., M. Plomp, A. J. Malkin, and P. Setlow.** 2008. Protozoal digestion of coat-defective *Bacillus subtilis* spores produces "rinds" composed of insoluble coat protein. Appl. Environ. Microbiol. **74:**5875–5881.
2. **Chada, V. G. R., E. A. Sanstad, R. Wang, and A. Driks.** 2003. Morphogenesis of *Bacillus* spore surfaces. J. Bacteriol. **185:**6255–6261.
3. **Driks, A.** 1999. *Bacillus subtilis* spore coat. Microbiol. Mol. Biol. Rev. **63:**1–20.
4. **Eichenberger, P., M. Fujita, S. T. Jensen, E. M. Conlon, D. Z. Rudner, S. T. Wang, C. Ferguson, K. Haga, T. Sato, J. S. Liu, and R. Losick.** 2004. The program of gene transcription for a single differentiating cell type during sporulation in *Bacillus subtilis*. PLoS Biol. **2:**1664–1683.
5. **Ghosh, S., B. Setlow, P. G. Wahome, A. E. Cowan, M. Plomp, A. J. Malkin, and P. Setlow.** 2008. Characterization of spores of *Bacillus subtilis* that lack most coat layers. J. Bacteriol. **190:**6741–6748.
6. **Henriques, A. O., and C. P. Moran, Jr.** 2007. Structure, assembly, and function of the spore surface layers. Annu. Rev. Microbiol. **61:**555–588.
7. **Hiragi, Y., K. Iijima, and H. Kadota.** 1967. Hexagonal single crystal pattern from the spore coat of *Bacillus subtilis*. Nature **215:**154–155.
8. **Hiragi, Y., and H. Kadota.** 1971. Fiber-structured material in a fraction of *Bacillus subtilis*. J. Bacteriol. **108:**579–581.
9. **Jones, C. A., N. L. Padula, and P. Setlow.** 2005. Effect of mechanical abrasion on the viability, disruption and germination of spores of *Bacillus subtilis*. J. Appl. Microbiol. **99:**1484–1494.
10. **Kadota, H., and K. Iijima.** 1965. The X-ray diffraction pattern of spores of *Bacillus subtilis*. Agric. Biol. Chem. **29:**80–81.
11. **Kadota, H., K. Iijima, and A. Uchida.** 1965. The presence of keratin-like substance in spore coat of *Bacillus subtilis*. Agric. Biol. Chem. **29:**870–875.
12. **Kim, H., M. Hahn, P. Grabowski, D. C. McPherson, M. M. Otte, R. Wang, C. C. Ferguson, P. Eichenberger, and A. Driks.** 2005. The *Bacillus subtilis* spore coat protein interaction network. Mol. Microbiol. **59:**487–502.
13. **Klobutcher, L. A., K. Ragkousi, and P. Setlow.** 2006. The *Bacillus subtilis* spore coat provides "eat resistance" during phagosomal predation by the protozoan *Tetrahymena thermophila*. Proc. Natl. Acad. Sci. USA **103:**165–170.
14. **Laaberki, M.-H., and J. Dworkin.** 2008. Role of spore coat proteins in the resistance of *Bacillus subtilis* spores to *Caenorhabditis elegans* predation. J. Bacteriol. **190:**6197–6203.
15. **Lee, K. S., D. Bumbaca, J. Kosman, P. Setlow, and M. J. Jedrzejas.** 2008. Structure of a protein-DNA complex essential for DNA protection in spores of *Bacillus* species. Proc. Natl. Acad. Sci. USA **105:**2806–2811.
16. **Mason, J. M., and P. Setlow.** 1986. Evidence for an essential role for small, acid-soluble spore proteins in the resistance of *Bacillus subtilis* spores to UV light. J. Bacteriol. **167:**174–178.
17. **Moir, A.** 1981. Germination properties of a spore coat-defective mutant of *Bacillus subtilis*. J. Bacteriol. **146:**1106–1116.
18. **Nicholson, W. L., and P. Setlow.** 1990. Sporulation, germination and outgrowth, p. 391–450. *In* C. R. Harwood and S. M. Cutting (ed.), Molecular biological methods for *Bacillus*. John Wiley and Sons, Chichester, United Kingdom.
19. **Paidhungat, M., B. Setlow, A. Driks, and P. Setlow.** 2000. Characterization of spores of *Bacillus subtilis* which lack dipicolinic acid. J. Baceriol. **182:**5505–5512.
20. **Palmer, K. J., R. Y. Wong, and J. C. Lewis.** 1972. The crystal structure of strontium dipicolinate tetrahydrate, $Sr \cdot C_7H_3NO_4 \cdot 4H_2O$. Acta Crystallogr. B **28:**223–228.
21. **Pogliano, K., E. Harry, and R. Losick.** 1995. Visualization of the subcellular location of sporulation proteins in *Bacillus subtilis* using immunofluorescence microscopy. Mol. Microbiol. **18:**459–470.
22. **Setlow, B., and P. Setlow.** 1996. Role of DNA repair in *Bacillus subtilis* spore resistance. J. Bacteriol. **178:**3486–3495.
23. **Setlow, P.** 2006. Spores of *Bacillus subtilis*: their resistance to radiation, heat and chemicals. J. Appl. Microbiol. **101:**514–525.

Downloaded from jb.asm.org at NAT INST OF HEALTH LIB on November 23, 2009